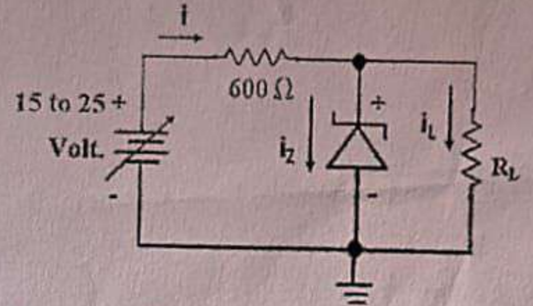
Institute of Engineering and Technology
Devi Ahilya University, Indore
Mid Sem Test II November 2023

Class/Section : BE (IT) A & B Section
Sub/Sub code : 1ETRC4- Basic Electronics

Max. Marks : 20
Time : 1 hour

Note :: Attempt only 4 questions. All questions carry equal marks.

Q.1) (a) The Zener voltage of the Zener diode shown in figure is 12V. Find the Maximum and Minimum Currents. [3]



(b) What are the differences between Zener breakdown and avalanche breakdown [2]

Q.2) Describe the process of electron-hole recombination and how it relates to light emission in an LED. Also discuss the semiconductor material that contributes to the LED's. [5]

Q.3) Explain fixed-biased with Emitter resistor and drive expression for operating point (V_{CE}, I_C). Why this circuit is thermally stable as compare self-biased without Emitter resistor. [5]

Q.4) Draw circuit of Voltage divider biased, How to resolve in to equivalent circuit. Also calculate expression for operating point (V_{CE}, I_C). [5]

Q.5) In the figure, assume that Vcc= 9V, R_b = 360 KΩ, Rc =2KΩ, β=100, V_{be}=0.7 V, Calculate Q point (V_{CE}, I_C). construct dc load line and mention Q point on load line. [5]